2.2. WORK FLOW OF THE PROJECT

PDB

1. PDB ID: 1M17 (Wild-Type EGFR)

Title: *Crystal structure of the epidermal growth factor receptor (EGFR) tyrosine kinase domain complexed with erlotinib (an inhibitor)*
Organism: *Homo sapiens (Human)*
Chain: A (monomeric form)
Resolution: 2.6 Å
Method: X-ray diffraction
Length: 312 amino acids (residues 696–1002, kinase domain)

2. PDB ID: 2ITY (L858R mutant EGFR)

Title: *Crystal structure of the L858R mutant of the human epidermal growth factor receptor (EGFR) kinase domain in complex with gefitinib (Iressa)*
Organism: *Homo sapiens (Human)*
Chain: A (single chain)
Resolution: 2.6 Å
Experimental Method: X-ray diffraction
Structure Type: Kinase domain of EGFR (residues 696–1022)
Length: 327 amino acids

PubChem

PubChem CID: 123631 (Gefitinib)
Gefitinib is a quinazoline-derived tyrosine kinase inhibitor (TKI) with molecular formula $C_{22}H_{24}ClFN_4O_3$ and molecular weight 446.9 g/mol. It functions as an **epidermal growth factor receptor (EGFR) antagonist**, blocking ATP binding pockets at the kinase domain to inhibit receptor autophosphorylation and downstream signaling, thereby preventing uncontrolled cell proliferation.
Download Format: 3D SDF format

Receptor Preparation Major Steps

1. The receptor (EGFR) and ligand (Gefitinib) structures were imported into PyRx and converted to .pdbqt format.
2. Water molecules and non-essential heteroatoms were removed to optimize docking accuracy.
3. Energy minimization before grid setup and docking.

Ligand Preparation Major Steps

i. Convert it to PDB format using Open Babel.
ii. Add hydrogens and assign Gasteiger charges.
iii. Define rotatable bonds for flexible docking.

TOOL/SOFTWARE USED
AutoDock Tools (ADT)

File 1- EGFR_wildtype.pdb

File 2- EGFR_mutant_2ITY.pdb

File 3- Gefitinib.pdb

The files are converted into the PDBQT Format

1. EGFR_wildtype.pdbqt
2. EGFR_mutant_2ITY.pdbqt
3. Gefitinib.pdbqt

Molecular Docking

i. Load receptor (wild/mutant EGFR) and ligand (Gefitinib) PDBQT files into PyRx.
ii. Visualize both molecules in the 3D viewer.
iii. Set the grid box around the ATP-binding pocket (using coordinates from the reference complex).
iv. Run AutoDock Vina to perform docking.
v. Record the binding affinities and RMSD values of the top poses.

Software Used- (PyRx)

Molecular Visualization (PyMOL)

i. The best docked complex from PyRx is imported into PyMOL for visualization.
ii. Receptor (wild or mutant EGFR) and ligand (Gefitinib) are loaded to view the binding pose.
iii. The complex is visualized in cartoon and surface formats to observe ligand fitting.
iv. Key interacting residues are identified to confirm stable and accurate binding at the ATP site.

Software Used- (PyMOL)